



Homework 2

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Problem 2.1

Proof. Contrapositive: if $r^{1/5}$ is rational, then r is rational. Assume that $r^{1/5}$ is rational. Then there exist $m, n \in \mathbb{N}$ with

$$r^{1/5} = \frac{m}{n}.$$

Therefore,

$$r = (r^{1/5})^5 = \left(\frac{m}{n}\right)^5,$$

which is rational. Thus, the contrapositive is true, and consequently, if r is irrational, then $r^{1/5}$ is irrational. $\square$

Problem 2.2

Prove by contradiction that $\log_3 18$ is irrational.

Proof.

$$\log_3 18 = \log_3 (2 \times 9) = \log_3 2 + \log_3 9 = 2 + \log_3 2$$

2 is clearly rational, we have to prove by contradiction that $\log_3 2$ is irrational because the addition of a rational and irrational number is irrational

$$\log_3 2 = \frac{m}{n} \quad n, m \in \mathbb{N}$$

$$n \log_3 2 = m$$

$$3^{n \log_3 2} = 3^m$$

$$2^n = 3^m$$

2 is even and an even number raised to the power of any natural number is always even and 3 is an odd number and an odd number raised to the power of any natural number is always odd so we have $odd = even$ which is false so by contradiction $\log_3 18$ is irrational $\square$

Problem 2.3

Identify exactly where the bugs are in the following bogus proof:

Bogus Claim: If a and b are two equal real numbers, then $a = 0$.

Proof:

$$a = b \tag{1}$$

$$a^2 = ab \tag{2}$$

$$a^2 - b^2 = ab - b^2 \tag{3}$$

$$(a - b)(a + b) = (a - b)b \tag{4}$$

$$a + b = b \tag{5}$$

$$a = 0 \tag{6}$$

Because $a = b \implies a - b = 0$ and dividing both sides by 0 on step 4 is illegal



Problem 2.4

$$\alpha \vee \beta \equiv \neg\alpha \rightarrow \beta$$

$$\alpha \wedge \beta \equiv \neg(\alpha \rightarrow \neg\beta)$$

$$(\alpha \vee \beta) \wedge (\beta \rightarrow \alpha) \equiv \neg((\alpha \vee \beta) \rightarrow \neg(\beta \rightarrow \alpha))$$

$$(\alpha \vee \beta) \wedge (\beta \rightarrow \alpha) \equiv \neg((\neg\alpha \rightarrow \beta) \rightarrow \neg(\beta \rightarrow \alpha))$$

Problem 2.5

Legend: T = true, F = false.

a) $((A_1 \wedge A_2) \rightarrow A_3) \rightarrow (A_1 \vee A_2)$

A_1	A_2	A_3	$A_1 \wedge A_2$	$(A_1 \wedge A_2) \rightarrow A_3$	$A_1 \vee A_2$	Result
T	T	T	T	T	T	T
T	T	F	T	F	T	T
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	F	T	T	T
F	T	F	F	T	T	T
F	F	T	F	T	F	F
F	F	F	F	T	F	F

b) $(\neg A_5 \rightarrow A_2) \rightarrow (A_5 \wedge A_2)$

A_5	A_2	$\neg A_5$	$\neg A_5 \rightarrow A_2$	$A_5 \wedge A_2$	Result
T	T	F	T	T	T
T	F	F	T	F	F
F	T	T	T	F	F
F	F	T	F	F	T



c) $(A_1 \wedge A_2 \rightarrow A_3 \wedge A_4) \longleftrightarrow (\neg(A_1 \vee A_3) \rightarrow \neg(A_2 \vee A_4))$

Precedence: $\neg > \wedge > \vee > \rightarrow > \leftrightarrow$, so the left side reads $((A_1 \wedge A_2) \rightarrow (A_3 \wedge A_4))$. Let **L** be the left side of $\leftrightarrow$ and **R** the right side.

A_1	A_2	A_3	A_4	$A_1 \wedge A_2$	$A_3 \wedge A_4$	L	$\neg(A_1 \vee A_3)$	$\neg(A_2 \vee A_4)$	R	L$\leftrightarrow$R
T	T	T	T	T	T	T	F	F	T	T
T	T	T	F	T	F	F	F	F	T	F
T	T	F	T	T	F	F	F	F	T	F
T	T	F	F	T	F	F	F	F	T	F
T	F	T	T	F	T	T	F	F	T	T
T	F	T	F	F	F	T	F	T	T	T
T	F	F	T	F	F	T	F	F	T	T
T	F	F	F	F	F	T	F	T	T	T
F	T	T	T	F	T	T	F	F	T	T
F	T	T	F	F	F	T	F	F	T	T
F	T	F	T	F	F	T	T	F	F	F
F	T	F	F	F	F	T	T	F	F	F
F	F	T	T	F	T	T	F	F	T	T
F	F	T	F	F	F	T	F	T	T	T
F	F	F	T	F	F	T	T	F	F	F
F	F	F	F	F	F	T	T	T	T	T

Problem 2.6

Since $(\beta \wedge \neg\beta)$ is a contradiction, its truth value is always false (F).

- If α is true (T), then $\alpha \rightarrow (\beta \wedge \neg\beta) \equiv T \rightarrow F \equiv F$.
- If α is false (F), then $\alpha \rightarrow (\beta \wedge \neg\beta) \equiv F \rightarrow F \equiv T$.

In both cases, $\alpha \rightarrow (\beta \wedge \neg\beta)$ yields the same truth value as $\neg\alpha$. Therefore, the two formulas are logically equivalent.

Problem 2.7

Show that $(\alpha \rightarrow \beta) \vee (\beta \rightarrow \alpha)$ is a tautology.

$(\alpha \rightarrow \beta) \vee (\beta \rightarrow \alpha)$ is false only when $(\alpha \rightarrow \beta)$ and $(\beta \rightarrow \alpha)$ are false. For $(\alpha \rightarrow \beta)$ to be false α should be true and β false but then $(\beta \rightarrow \alpha)$ would be true, and if $(\beta \rightarrow \alpha)$ is false then β should be true and α false but again $(\alpha \rightarrow \beta)$ would become true so we proved that $(\alpha \rightarrow \beta) \vee (\beta \rightarrow \alpha)$ is a tautology.



Problem 2.8

Using De Morgan's laws:

$$P \equiv (A \vee B) \rightarrow C \equiv \neg(A \vee B) \vee C \equiv (\neg A \wedge \neg B) \vee C$$

Distributing $\vee C$:

$$P \equiv (\neg A \vee C) \wedge (\neg B \vee C) \equiv (A \rightarrow C) \wedge (B \rightarrow C)$$

We simplify Q using contrapositives ($\neg C \rightarrow \neg X \equiv X \rightarrow C$):

$$Q \equiv (\neg C \rightarrow \neg A) \vee (\neg C \rightarrow \neg B) \equiv (A \rightarrow C) \vee (B \rightarrow C)$$

Since conjunction implies disjunction ($X \wedge Y \implies X \vee Y$), we have:

$$((A \rightarrow C) \wedge (B \rightarrow C)) \implies ((A \rightarrow C) \vee (B \rightarrow C))$$

Therefore, $P \rightarrow Q$ holds, but P and Q are not logically equivalent.

The converse fails: take $A = \text{F}$, $B = \text{T}$, $C = \text{F}$. Then $A \rightarrow C$ is true, so Q is true, but $B \rightarrow C$ is false, so P is false. Hence $Q \not\rightarrow P$ and $P \not\equiv Q$.

Answer: b) $P \rightarrow Q$.

Problem 2.9

- a) $D \rightarrow (B \wedge C)$
- b) $C \rightarrow (\neg A \wedge \neg B)$
- c) $D \leftrightarrow (C \wedge \neg A)$
- d) $D \rightarrow (\neg C \rightarrow A)$
- e) $(\neg D \rightarrow C) \wedge (D \rightarrow \neg B)$
- f) $A \rightarrow ((\neg B \wedge \neg C) \rightarrow D)$
- g) $((A \wedge B \wedge C) \leftrightarrow \neg D) \wedge ((\neg A \wedge \neg B) \rightarrow (D \rightarrow C))$

Problem 2.10

$$\begin{aligned}
 (\alpha \wedge \neg \beta) \rightarrow (\alpha \wedge \beta) &\equiv \neg(\alpha \wedge \neg \beta) \vee (\alpha \wedge \beta) && \text{(Conditional Law)} \\
 &\equiv (\neg \alpha \vee \beta) \vee (\alpha \wedge \beta) && \text{(De Morgan's Law)} \\
 &\equiv \neg \alpha \vee (\beta \vee (\beta \wedge \alpha)) && \text{(Associativity \& Commutativity)} \\
 &\equiv \neg \alpha \vee \beta && \text{(Absorption Law)}
 \end{aligned}$$

So $\neg \alpha \models (\alpha \wedge \neg \beta) \rightarrow (\alpha \wedge \beta)$ because $(\alpha \wedge \neg \beta) \rightarrow (\alpha \wedge \beta) \equiv \neg \alpha \vee \beta$ and $\neg \alpha \vee \beta$ is true if either $\neg \alpha$ or β is true and when $\neg \alpha$ is true $\neg \alpha \vee \beta$ is true regardless of β .



Problem 2.11

1. Proof of Contrapositive Rule

We know that $(\alpha \rightarrow \beta) \equiv (\neg\beta \rightarrow \neg\alpha)$ and using logical equivalences:

$$\begin{aligned}
\alpha \rightarrow \beta &\equiv \neg\alpha \vee \beta && \text{(Conditional Law)} \\
&\equiv \beta \vee \neg\alpha && \text{(Commutativity)} \\
&\equiv \neg(\neg\beta) \vee \neg\alpha && \text{(Double Negation)} \\
&\equiv \neg\beta \rightarrow \neg\alpha && \text{(Conditional Law)}
\end{aligned}$$

Because $\alpha \rightarrow \beta \equiv \neg\beta \rightarrow \neg\alpha$, the premise logically implies the conclusion:

$$(\alpha \rightarrow \beta) \models (\neg\beta \rightarrow \neg\alpha)$$

2. Proof of Composed Contrapositive Rule

$$\begin{aligned}
(\alpha \wedge \beta) \rightarrow \gamma &\equiv \neg(\alpha \wedge \beta) \vee \gamma && \text{(Conditional Law)} \\
&\equiv (\neg\alpha \vee \neg\beta) \vee \gamma && \text{(De Morgan's Law)} \\
&\equiv \neg\alpha \vee (\neg\beta \vee \gamma) && \text{(Associativity)} \\
&\equiv \neg\alpha \vee (\gamma \vee \neg\beta) && \text{(Commutativity)} \\
&\equiv (\neg\alpha \vee \gamma) \vee \neg\beta && \text{(Associativity)} \\
&\equiv \neg(\alpha \wedge \neg\gamma) \vee \neg\beta && \text{(De Morgan's Law: } \neg(A \wedge \neg B) \equiv \neg A \vee B) \\
&\equiv (\alpha \wedge \neg\gamma) \rightarrow \neg\beta && \text{(Conditional Law)}
\end{aligned}$$

Because $(\alpha \wedge \beta) \rightarrow \gamma \equiv (\alpha \wedge \neg\gamma) \rightarrow \neg\beta$, the deduction holds:

$$((\alpha \wedge \beta) \rightarrow \gamma) \models ((\alpha \wedge \neg\gamma) \rightarrow \neg\beta)$$

Problem 2.12

- a) **No.** Socrates does not state what should happen if he is not guilty
- b) **Yes.**
- c) **No.** He did not state explicitly that he should be punished only if he is guilty.

Problem 2.13

Let the propositions be defined as:

S : Smith is the murderer.

L : Jones is lying.

V : Jones has seen Smith last night.

M : The murder happened after midnight.



The premises are formalized as:

1. $P_1 : \neg V \rightarrow (S \vee L)$
2. $P_2 : \neg S \rightarrow (\neg V \wedge M)$
3. $P_3 : M \rightarrow (S \vee L)$

We test if $P_1 \wedge P_2 \wedge P_3 \models S$.

Assume S is false ($S \equiv \text{F}$). From P_2 , since $\neg S \equiv \text{T}$, we obtain:

$$\neg V \wedge M \equiv \text{T} \implies \neg V \equiv \text{T} \text{ and } M \equiv \text{T}.$$

Substituting $M \equiv \text{T}$ into P_3 gives $M \rightarrow (S \vee L) \equiv \text{T} \rightarrow (\text{F} \vee L) \equiv L$. Thus, for P_3 to be true, L must be true ($L \equiv \text{T}$).

Evaluating P_1 under this assignment ($\neg V \equiv \text{T}, S \equiv \text{F}, L \equiv \text{T}$):

$$\neg V \rightarrow (S \vee L) \equiv \text{T} \rightarrow (\text{F} \vee \text{T}) \equiv \text{T}.$$

All premises P_1, P_2, P_3 evaluate to T under the valuation ($S = \text{F}, L = \text{T}, V = \text{F}, M = \text{T}$).

Conclusion: The deduction is **invalid**. We cannot conclude that Smith is the murderer; we can only deduce that $S \vee L$ is true (either Smith is the murderer or Jones is lying).

Problem 2.14

Three boxes are presented. One contains gold, the other two are empty. Each box has a clue printed on it:

- Box 1: “The gold is not here”
- Box 2: “The gold is not here”
- Box 3: “The gold is in Box 2”

Only one clue is true; the other two are false. Which box has the gold?

Formalization

Let the propositional variables be:

- G_1 : the gold is in Box 1.
- G_2 : the gold is in Box 2.
- G_3 : the gold is in Box 3.

The clues written on the boxes translate to:

$$\begin{array}{ll} C_1 \equiv \neg G_1 & \text{(Box 1: “the gold is not here”)} \\ C_2 \equiv \neg G_2 & \text{(Box 2: “the gold is not here”)} \\ C_3 \equiv G_2 & \text{(Box 3: “the gold is in Box 2”)} \end{array}$$

The problem’s conditions are:

1. Exactly one of G_1, G_2, G_3 is true (the gold is in exactly one box).
2. Exactly one of C_1, C_2, C_3 is true (only one clue is true).

We must find the assignment of G_1, G_2, G_3 that satisfies both conditions.



Truth table

Legend: **T** = true, **F** = false.

G_1	G_2	G_3	$C_1 = \neg G_1$	$C_2 = \neg G_2$	$C_3 = G_2$	Exactly one box has gold	Exactly one clue is true	Valid
T	T	T	F	F	T	F	T	F
T	T	F	F	F	T	F	T	F
T	F	T	F	T	F	F	T	F
T	F	F	F	T	F	T	T	T
F	T	T	T	F	T	F	F	F
F	T	F	T	F	T	T	F	F
F	F	T	T	T	F	T	F	F
F	F	F	T	T	F	F	F	F

Solution

Scanning the table, the **Valid** column is true in exactly one row: $G_1 = T$, $G_2 = F$, $G_3 = F$. In that row, C_1 is false, C_2 is true, and C_3 is false, so exactly one clue (Box 2's) is true, as required, and exactly one box (G_1) contains the gold.

The gold is in Box 1. ■

Problem 2.15

- Negation:** "There exists a prime number that is even"
- Negation:** "There exists a computer program P such that P compiles without error messages and P is not correct."
- Contrapositive:** "If a set is uncountable, then it is infinite."
- Contrapositive:** "If p is even, then p is not a prime number."
- Converse:** "If you pass the exam, then you study hard."
- Negation:** "All people study Discrete Math" (or "Everyone studies Discrete Math").
- Negation:** $\exists x \in \mathbb{R}, \forall y \in \mathbb{Z}, (x + y < 0)$.

Proof: Let $x \in \mathbb{R}$ be arbitrary. Since the integers are unbounded above, there exists an integer y such that

$$y \geq -x.$$

(For example $y = \lceil -x \rceil$ because the ceiling of $-x$, which is always an integer.)

Adding x to both sides of $y \geq -x$ gives

$$x + y \geq x + (-x) = 0.$$

Since $x \in \mathbb{R}$ was arbitrary, and for each such x we exhibited an integer y satisfying $x + y \geq 0$, it follows that

$$\forall x \in \mathbb{R} \exists y \in \mathbb{Z} \text{ such that } (x + y) \geq 0. \quad \blacksquare$$

- Negation:** $\exists x \in \mathbb{R}$ such that $x(x + 1) > 0$ and $-1 \leq x \leq 0$.